Table 3

Operational performance of future-attack forecasting models at fixed decision thresholds

Horizon	Model	Threshold	Window-level performance			Window errors		Episode-level alerts
			Precision (%)	Recall (%)	FPR (%)	FP	FN	Benign episodes alerted
H5	CR-LSTM	0.20	52.4	87.9	5.08	899	136	0/75 (0.0%)
		0.50	68.8	60.9	1.76	311	440	0/75 (0.0%)
	CR-TCN	0.20	51.3	89.2	5.39	954	122	3/75 (4.0%)
		0.50	67.3	67.7	2.09	370	363	3/75 (4.0%)
H10	CR-LSTM	0.20	71.5	92.8	5.02	832	161	3/75 (4.0%)
		0.50	80.9	81.1	2.60	431	426	0/75 (0.0%)
	CR-TCN	0.20	68.7	94.1	5.82	966	133	5/75 (6.7%)
		0.50	79.7	83.5	2.88	477	372	3/75 (4.0%)
H15	CR-LSTM	0.20	80.9	90.5	4.67	722	321	12/75 (16.0%)
		0.50	89.7	83.2	2.08	321	567	4/75 (5.3%)
	CR-TCN	0.20	79.4	91.5	5.19	803	288	13/75 (17.3%)
		0.50	88.1	84.1	2.49	385	536	5/75 (6.7%)

Notes: Precision and recall use true-positive attack windows. FPR = false positives / all true-negative windows; FP and FN are window counts.
Benign episodes alerted = number of entirely benign episodes with at least one alert / 75; this is not the same quantity as window-level FPR.
Synthetic NAF v2 development validation, 18,839 overlapping decision windows. Thresholds 0.20 and 0.50 are illustrative operating points, not optimized deployment settings.
The original audit also contains threshold 0.10; it remains preserved in Research_Outputs.